Cloud data warehouse

00:00Thanks for joining me as we venture into the wide world of cloud data warehousing!

00:05There's so much data out there, it's truly dizzying.

00:08So it's no surprise that businesses have struggled to figure out where to keep it all.

00:13The fact is, traditional databases struggled to keep up with the evolving demands of data analytics.

00:19Luckily, cloud data warehouses are emerging to fill the need.

00:24How do they do it?

00:25Well, that's what we're going to learn about in this video.

00:28First, a cloud data warehouse is a large-scale, data storage solution hosted on remote servers by a cloud service provider.

00:36To understand this better, picture it like a huge warehouse where large amounts of different types of containers from various places are stored.

00:44The cloud data warehouse can collect, store, integrate, and analyze data.

00:50There are many advantages to this structure.

00:53Cloud data warehouses are typically fully managed by the cloud provider.

00:58This means that the cloud provider takes care of various operational tasks and maintenance, allowing

01:03users to focus on utilizing the data and gathering insights rather than handling the underlying infrastructure.

01:11This saves time, money, and resources.

01:15Cloud data warehouses also have more uptime compared to on-premises data warehouses.

01:22Uptime is the amount of time a machine is operational.

01:24And, of course, only working computers have the ability to scale and support increased demands for data.

01:32Next, cloud warehouses can integrate separated data by gathering data from various structured sources within

01:37an organization, like sales systems, email lists, and websites, and pulling it all into one place.

01:46This integrated data then can be analyzed for some pretty exciting and useful business insights.

01:52Another big advantage is that cloud data warehouses provide real-time analytics, ensuring users have quick access to the latest information.

02:01And in business, being fast is usually the key to outperforming the competition.

02:06Cloud data warehouses also offer some really cool artificial intelligence, or AI, and machine learning, or ML, capabilities.

02:14And when you apply AI and ML to your data analysis, this really powers up the possibilities.

02:20My team worked on a recent project where we built a predictive model to help Google

02:23anticipate demand for office amenities, such as its cafés, and help save money and reduce waste.

02:32Using ML tools, we were able to test over 30 factors across months and months of data and

02:37build a model that could forecast demand with enough accuracy and time to allow on-the-ground teams to adjust accordingly.

02:46Pretty cool, right?

02:48Last but not least, cloud data warehouses enable custom reporting and analysis.

02:54This means that users can analyze and generate reports specifically from historical data because it

02:58is stored on a separate server from data related to current business transactions and day-to-day operations.

03:06As you've probably figured out, the types and amounts of data that companies need to organize are only growing, which means so is the demand for data storage.

[03:15](03:15)Luckily, cloud data warehouses are up for the challenge with the added benefits of management and analysis to make it even easier to use the data you have.